

Matthew Cai

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Education

University of California, Berkeley

Bachelor of Science, Electrical Engineering and Computer Science

Expected December 2027

GPA: 4.00/4.00

Undergrad Coursework: SICP, DSA, Machine Structures, Circuits, Computer Architecture, Computer Security, Compilers, OS, Advanced Algorithms, Databases, Machine Learning, Reinforcement Learning, Linear Algebra, Discrete Math, Calculus I/II/III, Abstract Algebra

Graduate Coursework: Computer Architecture (audit), Computer Security, Performance Optimization

Leadership: Berke1337 Cybersecurity Club, Quantum Computing at Berkeley (QCB), Eta Kappa Nu (HKN)

Experience

Tesla | (Incoming) Product Security Engineering Intern, Vehicle Software

August 2026 – December 2026

Meta | Offensive Security Engineering Intern

June 2026 – August 2026

Meta | Production Engineering (Network Security) Intern

May 2025 – August 2025

- Redesigned concurrency and pipelining for validation of network access control list (ACL) generation, reducing runtime by **99%**
- Devised a scalable, custom lexer to automate generating **6,000+** lines of AST grammars from Apache Thrift interfaces
- Replaced quadratic-complexity IP prefix matching with radix trees in TACACS+ validator, cutting runtime from **2.5 min to 1 sec**
- Developed CLI to enhance and debug ACL generation library capirca by leveraging real-time OLAP databases and entity graphs

MITRE | Reverse Engineering Intern

June 2024 – August 2024

- Assessed the performance of Ghidra BSim, an open-source binary differential analysis tool, across diverse architectures and OSS
- Integrated Ghidra's Python API into a headless CLI with parallelized BSim database queries and statistical results for engineers
- Reverse engineered firmware vulnerabilities and exploited side channels in an embedded ARM device for a CTF, tied for **1st place**

Research

Graduate Computer Security Research at UC Berkeley (CS 261) | Security Researcher [1]

February 2026 – May 2026

- Designed the *first* LLM-based decompiler for **Go source recovery**, integrating Gemini, Ghidra, GoReSym, and custom Go heuristics
- Curated *first* Go decompiler benchmark of **1.8M functions** extracted from 1600 unique binaries across **120 OSS** projects
- Introduced novel evaluation metrics: CodeBLEU, LLM-as-a-judge, and syntax validation scoring weighted by code and AST complexity

Security Group at UC Berkeley (D. Song) | Security Researcher [2]

January 2025 – December 2025

- Engineered sandboxed LLM-agent MCP for automated PoC generation across **1,500 vulnerabilities** in **200 OSS** projects
- Discovered **15 zero-days** during benchmarking, leading to CVE-2025-7797, CVE-2025-7464, and CVE-2025-7462
- Architected framework to evaluate LLM agents on generating ROP chains for real-world binaries, e.g. GNU Coreutils

Industrial Cyber-Physical Systems (iCyPhy) at UC Berkeley | Systems Researcher [3]

October 2025 – December 2025

- Extended the GameTime algorithm for worst-case execution time (WCET) analysis of the Lingua Franca polyglot language
- Programmed 3 FRDM-K64F microcontrollers with Zephyr RTOS to demo WCET analysis on a real embedded, distributed system

Publications

[1] **ComeBack: Assessing LLM Decompilation for Go**

Preprint, 2026

Matthew Cai, Jonah Bedouch

[2] **CyberGym: Evaluating AI Agents' Real-World Cybersecurity Capabilities at Scale**

ICLR Oral 2026

Zhun Wang, Tianneng Shi, Jingxuan He, Matthew Cai, Jialin Zhang, Dawn Song

[3] **Formally Verified WCET Analysis of Distributed Lingua Franca Systems via GameTime**

Preprint, 2025

Zeynep Guzey, Anagha Ananth, Matthew Cai, Jiaquan Li, Shaokai Lin

Activities and Projects

Cybersecurity Capture the Flag | Vulnerability Researcher

September 2023 – Present

- Member of **top 3** US CTF Team, *Squid Proxy Lovers*: 3rd at Google 2025, 2nd at idek 2025, 2nd at Dice Qual 2025, 5th at Lake 2025
- Skills: symmetric cryptanalysis; lattice attacks; stack/heap userspace ELF exploitation; microarchitecture exploitation

Berke1337 Cybersecurity Club | Officer

August 2024 – Present

- Helped build custom homelab for network defense training and created custom pwn challenges for demonstrations and practice
- Organized training and recruitment for the club CTF team; taught members software exploitation, fuzzing, cryptography, etc.

Google × Flare Verifiable AI Hackathon | 3rd Place

March 2025

American Invitational Mathematics Examination | 5-time qualifier, highest score of 7

2019 – 2024

Skills

Languages: C/C++, Python, SageMath, Go, Rust, OCaml, Lisp, Java, SQL, x86, ARM, RISC-V, Bash, Chisel, LaTeX

Tools: Linux, Git/Mercurial, Docker, Make, GDB/pwndbg, Wireshark, Ghidra/Binary Ninja/IDA, VirtualBox

Research Interests: security, microarchitecture, performance, databases, distributed systems, AI safety, ML systems